Probability Risk Assessment of Solar Energy Events Across Space Mission Environments

Executive Summary

This report presents a probabilistic risk assessment (PRA) of radiation exposure across three progressively more challenging mission environments: near-space high-altitude flight, Low Earth Orbit (LEO), and lunar surface operations.

The analysis demonstrates that radiation risk is not solely determined by deterministic approaches, but is strongly influenced by low-probability, high-consequence solar energetic particle events.

Key findings include:

- Radiation risk profiles differ significantly across high altitude, LEO, and lunar mission environments due to variations in shielding and particle exposure conditions.
- PRA-based approaches provide additional insight into uncertainty and extreme-event contributions beyond conventional deterministic methods.
- The analyzed LEO and lunar case studies demonstrate how environmental shielding conditions influence the relative contribution of SEP events to total radiation exposure.
- Radiation risk assessment benefits from considering both continuous background exposure and transient solar particle events within an integrated probabilistic framework.

The objective of this report is not only to quantify radiation exposure, but to demonstrate how probabilistic radiation modeling can support architectural and operational decision-making in space systems.

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1. Introduction

Radiation exposure is a critical consideration in human spaceflight and high-altitude operations. Traditional deterministic approaches often rely on conservative assumptions and cumulative exposure estimates, which may not fully capture the stochastic nature of solar particle events.

This report applies a probabilistic risk assessment (PRA) framework to evaluate radiation risk across three mission environments:

- Near-space high-altitude flight (~30 km)
- Low Earth Orbit (LEO)
- Lunar surface operations

The study focuses on how low-probability, high-consequence radiation events alter mission-level risk and influence engineering design decisions.

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2. Case Study 1: Near-Space High-Altitude Flight (~30 km)

2.1 Objective

This case study uses the data from the GLE model by Rees (2023, 2024): a pressurized observation capsule on a stratospheric ballon flying from Sutherland Spaceport into the North Sea, with high latitude (~58–60° N), which represents a pessimistic high latitude flight during a GLE event.

2.2 Radiation Environment

Near-space high-altitude missions operating at approximately 30 km experience substantially elevated radiation exposure compared to conventional commercial aviation environments.

At this altitude, atmospheric shielding is significantly reduced, resulting in increased exposure to Galactic Cosmic Rays (GCR) and secondary particle cascades generated in the atmosphere. Under nominal conditions, GCR represents the dominant source of continuous background radiation exposure in the near-space environment.

In addition to continuous background exposure, high-altitude missions may also experience transient radiation increases associated with Ground Level Enhancement (GLE) events. GLEs occur when high-energy Solar Energetic Particles (SEPs), typically produced during major solar eruptions, penetrate Earth's magnetosphere and atmosphere sufficiently to produce enhanced radiation levels detectable at ground level.

Although rare, strong GLE events can generate short-duration but significant increases in radiation dose, particularly for high-latitude and high-altitude missions where geomagnetic shielding is reduced.

As a result, radiation risk in near-space flight environments consists of both:

- Continuous background exposure from GCR
- Low-probability, high-consequence exposure from GLE events

This case study focuses primarily on the probabilistic contribution of GLE events during the reference 6-hour mission.

2.3 Mathematical Formulation

GLEs are categorised into four classes (Weak, Strong, Severe, Extreme) (Rees 2023) using event-integrated intensity relative to the 1956 GLE 05, with associated maximum effective dose estimates for a high latitude 30 km flight.

To convert annual GLE frequencies into mission-specific likelihoods, each GLE class is modelled as a Poisson process.

For a representative near-space exposure duration of 6 hours, the probability of at least one GLE of class j occurring during the flight is calculated as

$$P_j = 1 - \exp(-\lambda_j T),$$

Where:

- λ_j is the annual frequency of the event class j, estimated as a best-estimate value from the historical observational record using a Poisson maximum likelihood estimator i.e. $\lambda_{Weak} = \frac{1}{10yr} = 0.1/year$.
- T is the flight duration expressed in years $T = \frac{6}{8760} =\times 10^{-4} yr$.

For very rare event, $\lambda T \ll 1$, $P_j \approx \lambda_j T$.

The expected effective dose contribution from GLEs for a single flight is then obtained by summing the probability-weighted dose consequences across all event classes:

$$E[D_{\mathrm{GLE}}] = \sum_j P_j \; D_j.$$

This metric represents the expected radiation dose per flight attributable to space weather. It accounts for the combined influence of both event likelihood and severity.

A few assumptions had been made with this Poisson process PRA model:

- The GLE event occurs randomly and independently.
- Assume the GLE event is quasi-steady over long-term. It does not imply solar activity is physically constant.

2.4 Results and Analysis

The Results are presented in Table 1. By adding the expected dose contribution from all GLE classifications, the expected effective dose contribution of the reference flight equals $E[D_{\mathrm{GLE}}] = 3.54 \times 10^{-4} mSv$.

GLE class	Annual frequency λ (1/yr)	Per-flight probability P (T=6 h)	Estimated max effective dose (mSv)	Expected dose contribution P*D (mSv / flight)
Weak	1.0×10^{-1}	6.85×10^{-5}	0.06	4.11×10^{-6}
Strong	2.0×10^{-2}	1.37×10^{-5}	4	5.48×10^{-5}
Severe	1.0×10^{-2}	6.85×10^{-6}	21	1.44×10^{-4}
Extreme	8.33×10^{-4}	5.71×10^{-7}	620	3.54×10^{-4}

Table 1 : Probabilistic radiation risk assessment for GLE scenarios during a representative 30 km near-space flight.

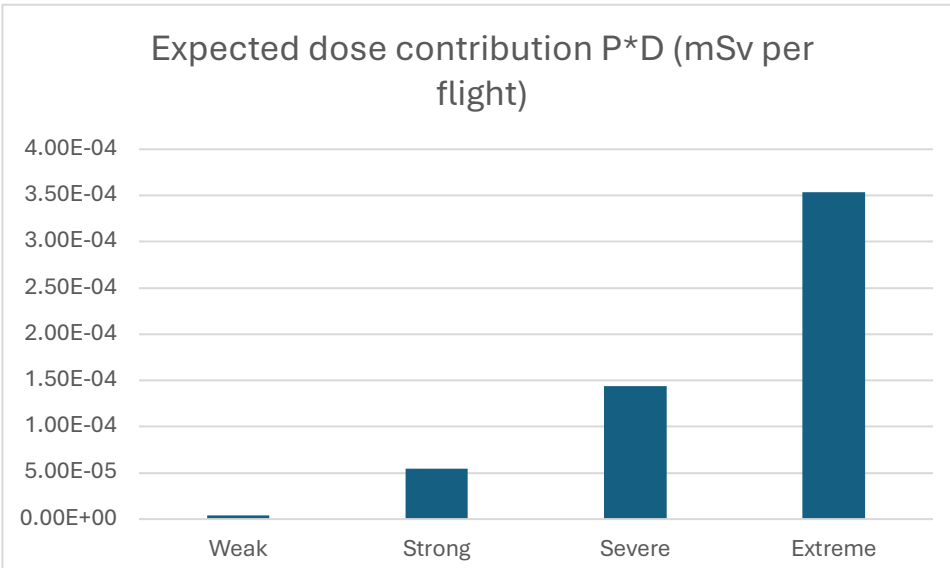

Figure 1: Comparison of expected dose contribution from each GLE Class

As illustrated in Figure 1, the expected dose contribution demonstrates a low probability, high consequence risk structure. While weak and strong GLEs occur more frequently, their contribution to the expected dose is small. In contrast, severe and extreme events are categorized as extremely rare occurrences. However, they exert a dominant influence on the expected value due to the high magnitude of their radiation doses.

Table 2 illustrates the probability that the effective dose exceeds selective dose thresholds. The exceedance probability $P(D > d)$ is determined for each dose threshold d. This value is calculated by summing the per-flight probabilities of all GLE classes with an estimated maximum effective dose above that threshold.

Dose threshold, d (mSv)	Exceeding GLE classes	P(D>d) per flight	Approx. 1 in N flights
1	Strong, Severe, Extreme	8.90×10^{-5}	~47,000
20	Severe, Extreme	7.42×10^{-6}	~135,000
100	Extreme only	5.71×10^{-7}	~1.75 million

Table 2: Probability of exceeding selected effective dose thresholds due to GLE events during a near-space flight (~30km, 6 hour)

The probabilistic results from Table 2 indicate that exceedance of 20 mSv is possible only under severe or extreme GLE scenarios and occurs with very low per-flight probability. Exceedance of 100 mSv occurs exclusively in the extreme GLE scenario, with an estimated probability of approximately 1 in 1.75 million flights, placing it firmly within the low-probability, high-consequence regime.

2.5 Interpretation and Design Implications

The PRA results support the implementation of targeted operational and architectural controls that address specific high-risk scenarios, rather than relying on conservative design measures applied uniformly under nominal conditions.

With the consideration of GLE events, four primary factors can be incorporated into the architecture design. First, mission profiles should include restrictions on altitude and latitude relative to the geomagnetic field to maximize shielding during flight. Second, continuous monitoring of solar activity is essential to facilitate the delay or cancellation of missions during periods of elevated solar risk. Third, supplemental shielding should be integrated into critical areas rather than applying uniform shielding across the entire structure, which reduces both mass and cost. Fourth, if a GLE occurs during mid-flight, operational protocols must ensure that all crew members move to a designated shielded area temporarily.

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3. Case Study 2: Low Earth Orbit (LEO)

3.1 Objective

This case study examines radiation risk for a short duration crewed mission in low-Earth orbit (LEO) in the presence of a significant solar energetic particle (SEP) event. The reference trajectory is a circular orbit at a 400 km altitude with a 51.6° inclination. The nominal mission length is 5 days.

3.2 Radiation Environment

At orbital altitudes, the Earth's magnetic protection is significantly reduced, leading to two primary changes:

- Increased Space Radiation: There is a much higher flow of GCR because there is less atmospheric and magnetic shielding to block them. This allows higher energy GCRs to penetrate the environment more effectively.
- Trapped Radiation: LEO missions must pass through the Van Allen belts. Specifically, spacecraft regularly transit the South Atlantic Anomaly (SAA), a region where the inner belt dips lower, creating an area of much more intense radiation.

While the geomagnetic field provides a baseline level of protection, it is not an absolute barrier during intense solar activity. During strong SEP events, the flux of high-energy protons can increase by several orders of magnitude, effectively overwhelming the nominal protection provided by the magnetosphere.

3.3 Radiation Environment Modeling

The baseline LEO radiation environment was modeled in OLTARIS. This model accounts for the primary contributions of GCR and trapped protons. Specifically, GCR fluxes were computed using the Badhwar–O'Neill 2020 model, and trapped protons radiation contributions were obtained from the AP9 model suite. Shielding was represented using slab geometries of 10 g/cm². The anatomical model was CAF with "Avg. US Population" weighted tissue.

To analyze the impact of a significant SEP event in the LEO environment, this study references in-situ dosimetry from the March 7 to 8, 2012 event measured aboard the International Space Station (ISS). The March 2012 SEP event is not considered an extreme historical event (e.g., not comparable to 1956 or 1989), but it represents a strong, realistic event that contemporary missions should be prepared to handle. The intensity of the SEP was stated by Fino et al. (2014) as "The flux is enhanced up to 10 times with respect to "quiet Sun" conditions." Semkova et al. (2014) reported radiation measurements from the Liulin-5 silicon telescope installed inside the MATROSHKA-R tissue-equivalent phantom on the ISS, which serves as a human-body analogue for assessing internal exposure. During the March 2012 SEP, Liulin-5 recorded a clear enhancement in dose rate above background at a depth of 40 mm within the phantom. The measured results indicated an additional effective dose equivalent of approximately 0.448 mSv attributable to the event.

3.4 Results and Analysis

The accumulated effective dose equivalents of GCR and trapped protons obtained from OLTARIS for a 5-day LEO mission are summarized in the table below:

Source	Dose (mSv)
GCR	0.734
Trapped Protons	1.096
Total	1.859
SEP Contribution	0.448

The additional accumulated effective dose equivalent from the reference SEP event is approximately 24% relative to the background dose from GCR and trapped protons. The ~24% increase in cumulative dose indicates that SEP contributions are non-negligible and can meaningfully influence mission-level exposure assessments.

3.5 Interpretation and Design Implication

The SEP event considered in this case study is a strong event based on in-flight measurements. More severe events would result in even higher dose rates and greater short-duration exposure. For this reason, SEP-driven risk should be treated as a time-critical hazard.

For LEO missions, orbital parameters are largely fixed by mission and program constraints. Radiation risk mitigation must therefore be addressed through architectural design within a defined orbital environment. Measurable quantities, such as dose rate and event-integrated dose, can be linked to specific design zones with enhanced shielding to support crew protection during elevated radiation conditions. Defining these relationships at the architectural level ensures that essential capabilities, such as accessible shielded locations, radiation monitoring systems, and operational responsiveness, are incorporated from the outset.

Continuous radiation monitoring, integrated with space weather data, should be treated as a core system capability to support real-time assessment against defined dose rate thresholds. By embedding this metric-driven decision logic into the design, radiation risk can be managed in a structured and efficient manner rather than being addressed through late-stage evaluation.

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4. Case Study 3: Lunar Surface Mission

4.1 Objective

Case study 3 focuses on the relative contributions of GCR and extreme SEP events to mission integrated dose, and evaluates how rare high fluence SEP events reshape the radiation risk structure despite their low occurrence probabilities.

The reference scenario is a 14-day stay inside a small pressurised lunar habitat. Only the surface phase is considered because this is the period when the crew is fully outside Earth's geomagnetic shielding. Launch, low Earth orbit, and cislunar transit exposures are not included.

4.2 Radiation Environment

Exposure on the lunar surface consists of a steady GCR background and intermittent SEP events. There is no lunar atmosphere or magnetic field, so exposure is determined entirely by interplanetary radiation conditions.

4.3 Radiation Environment Modeling

The occupied region of the habitat is modeled as a single shielding layer with an aluminum-equivalent thickness of 10 g/cm², and storm shelter shielding is not considered in this case study. The occurrence probability of large solar particle events is characterized using the long-term F_{30} statistics of Usoskin and Kovaltsov (2012).

For each event category a representative spectrum is transported through the 10 g/cm² shielding to obtain an event integrated dose D_j. Mission specific probabilities P_j follow a Poisson model with duration T.

The SEP spectra is specified using the Exponential in energy form

$$\Phi(> E) = J_0 \exp\left[-(E - E_1)/E_0\right. \quad \text{Eq. (3-1)}$$

Where:

- E_0 is the e-folding energy, which controls the hardness of the spectrum. $E_0 = 30\text{MeV}$, as it falls within the energy range that contributes most strongly to the event-integrated fluence in large SEP events.
- E_1 is the low energy cutoff. $E_1 = 10\text{MeV}$ to match the standard lower threshold of space-era fluence channels (F_{10}, F_{30}, etc.).
- J_0 is the normalization constant chosen so that the spectrum reproduces the target event fluence (e.g., F_{30}).

The extreme SEP fluence statistics are based on the cosmogenic isotope dataset of Usoskin and Kovaltsov (2012), which reconstructs the occurrence of large SEP events over more than 11,000 years. Their analysis provides return periods for key fluence thresholds defined using F30 (proton fluence above 30 MeV). key metric:

- F_{30}: Event-integrated proton fluence for energies > 30 MeV.

Usoskin and Kovaltsov (2012) identify extreme SEP event classes with proton fluence above 30 MeV corresponding to approximately 10, 20–30, and 50 times the SPE56 reference event (Table 3). Using these F_{30} values as $\Phi(>E)$ in Eq. (3-1), J_0 is obtained by solving the spectral integral for each case. The resulting J_0 values are listed in the fourth column of Table 3. The occurrence of

extreme SEP events is modelled using the same Poisson framework introduced in Case Study 1. For each event class j, the per-mission probability P_j is obtained by applying its annual frequency λ_j over 14-day mission interval $T = 0.0383$yr.

F_{30} (cm^{-2})	Annual probability (λ_j)	Frequency in years	J_0 (MeV^{-1})
1×10^{10} (10SPE56)	10^{-2} yr^{-1}	~100 years	0.65×10^9
2–3× 10^{10} (20-30SPE56)	10^{-3} yr^{-1}	~1,000 years	1.63×10^9
5×10^{10} (50SPE56)	10^{-4} yr^{-1}	~10,000 years	3.25×10^9

Table 3: Extreme SEP occurrence probabilities from Usoskin & Kovaltsov (2012)

The F30 for the 20 to 30 SPE56 class spans 2 to 3×10^{10} cm^{-2} (Table 3). In the absence of evidence for a strong skew, the midpoint value 2.5×10^{10} cm^{-2} is utilized as a representative value to compute J_0.

Radiation environment estimates for the lunar surface mission were generated using NASA's OLTARIS platform. GCR fluxes were computed using the Badhwar–O'Neill 2020 model. Shielding was represented using slab geometries of 10 g/cm². Anatomical model is CAF with "Avg. US Population" weighted tissue. GCR exposure is evaluated in 2010 solar minimum condition. Solar particle event (SEP) spectra are defined using a user-specified energy distribution, with J_0 values for each event category listed in Table 3, and fixed parameters $E_0 = 30$ MeV and $E_1 = 10$ MeV.

4.4 Results and Analysis

The results generated using the Badhwar–O'Neill 2020 model indicate that GCR contributes an effective dose of 9.358 mSv. Table 4 indicates that the representative SEP classes yield event-integrated doses substantially exceed the nominal GCR contribution. However, these large consequences are associated with correspondingly low occurrence frequencies.

SEP Class	Annual frequency λ (1/yr)	Per-mission probability P (T=14d)	Effective dose SEP event (mSv)	Expected dose contribution P*D (mSv per mission)
10SPE56	10^{-2}	3.83×10^{-4}	127	4.86×10^{-2}
20–30SPE56	10^{-3}	3.83×10^{-5}	317	1.21×10^{-2}
50SPE56	10^{-4}	3.83×10^{-6}	634	2.43×10^{-3}

Table 4: PRA of extreme SEP event to a 14-day Lunar surface mission.

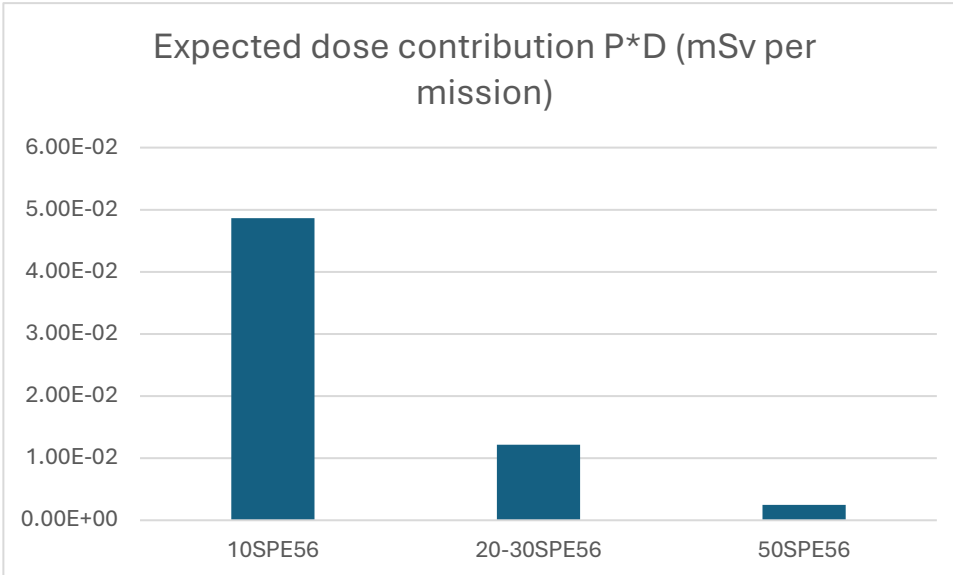

Figure 2: Comparison of expected dose contribution from each SPE Class

Figure 2 illustrates the expected SEP dose contributions for the three representative SPE classes. The pattern shows a steep decline in contribution as the events become less frequent but more intense.

From Table 4, expected SEP effective contribution per mission:

$$E[D_{\text{SEP}}] = \sum_j P_j \; D_{j,\text{SEP}} = 6.32 \times 10^{-2} mSv$$

Therefore, the total expected dose:

$$E[D_{\text{total}}] = D_{\text{GCR}} + E[D_{\text{SEP}}] = 9.42 mSv$$

Although extreme SEP events can produce doses substantially exceed the GCR contribution, their per-mission probabilities are very small. Therefore, extreme SEP events should be treated as a separate design condition alongside the continuous background.

4.5 Interpretation and Design Implications

Similar to LEO missions, radiation risk in lunar surface missions is incorporated in this framework through metric-driven design, where exposure quantities such as effective dose, dose rate, and event-integrated dose are linked to defined protection strategies.

Unlike the LEO environment, the lunar surface lacks geomagnetic and atmospheric shielding, leaving exposure levels determined entirely by the external radiation field. Therefore, radiation protection relies more heavily on the inherent shielding capability of the architecture rather than operational flexibility. While continuous exposure drives the baseline habitat design, severe SEP events define the protection requirements for contingency conditions. By linking exposure metrics to both baseline and contingency requirements, radiation risk is incorporated directly into the architectural design, ensuring the system addresses both cumulative exposure and extreme

events

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5. Conclusion

The case studies across different space environments show that radiation risk cannot be fully understood using deterministic approaches alone. Instead, the low-probability, high-consequence nature of SEP-driven radiation risk is better captured through probabilistic risk assessment (PRA) modeling.

The results demonstrate that probabilistic radiation modeling can support:

- Early-stage architectural design
- Operational planning
- Shielding strategy development
- Risk-informed mission decision-making

Radiation should therefore be treated not only as an environmental condition, but also as a system-level design driver.

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6. References

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